An Intelligent Cloud Resource and Cost Optimization Framework using Machine Learning and Explainable AI

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Abstract* — Modern cloud-native setups frequently experience significant financial inefficiencies due to over-provisioned resources, idle virtual machines, unattached storage volumes, out-of-date snapshots, and ineffective cloud cost management. Businesses are increasingly at danger from economic security threats, such as denial-of-wallet (DoW) attacks, in which hostile workloads intentionally demand enormous amounts of cloud resources, resulting in unnecessary expenditures and operational waste. Present-day FinOps solutions mostly provide rule-based monitoring and historical billing visualization; they have limited capabilities for predictive analytics, business-aware decision support, explainable recommendations, and secure cloud remedy. This project suggests CloudSense AI: An Intelligent Cloud Resource and Cost Optimization Framework using machine learning, explainable artificial intelligence (XAI), model context protocol (MCP), and the AWS SDK (Boto3). It combines Isolation Forest for cloud anomaly and denial-of-wallet detection, XGBoost for business-aware Unit Economics prediction, and Prophet for cloud cost forecasting. Explainable AI provides precise recommendations with confidence scores, estimated savings, risk assessment, feature importance, and root cause analysis to assist engineering decisions. The framework consists of an MCP for standardized communication with enterprise cloud services to reduce human operational effort and an Intelligent Cloud Remediation Engine that employs the AWS SDK (Boto3) for the safe execution of permitted cloud optimization actions. Instead of building Infrastructure-as-Code templates, the framework leverages secure API-based automation to perform controlled remediation tasks, such as EC2, EBS, and Auto Scaling resources. Every repair activity uses a Human-in-the-Loop (HITL) protocol, and administrator approval is needed before it can be carried out. The framework also provides real-time notifications using custom webhooks, Microsoft Teams, Discord, Slack, and email. An AI-powered conversational assistant allows users to interact with the cloud infrastructure using natural language and receive contextual explanations of AI recommendations. The proposed framework is expected to improve cloud resource utilization, detect cloud cost anomalies and denial-of-wallet attacks, increase operational transparency, financial efficiency, and decision-making in enterprise cloud environments, and strengthen cloud governance through secure AI-assisted remediation.

***Keywords* — Machine Learning (ML), Explainable Artificial Intelligence (XAI), Cloud Computing, Cloud Resource Optimization, Cloud Cost Optimization, FinOps, Model Context Protocol (MCP), AWS SDK (Boto3), Intelligent Cloud Remediation, Unit Economics, Human-in-the-Loop (HITL), Denial-of-Wallet (DoW) Detection.**

I. INTRODUCTION

Cloud computing has revolutionized contemporary information technology by offering scalable, adaptable, and affordable computing resources on a pay-as-you-go basis. [1] Despite these benefits, over-provisioned resources, idle virtual machines, ineffective scaling practices, and erratic workload demands frequently result in increased cloud expenditures for businesses. [2], [3] Cloud cost optimization is mainly reactive because current Financial Operations (FinOps) [4] solutions mainly rely on historical billing records and rule-based recommendations. [5], [6] Additionally, new financial risks, such as denial-of-wallet (DoW) [7] assaults, take advantage of cloud elasticity to create exorbitant operating costs that are difficult to identify using traditional monitoring techniques. [8] [9] This study suggests CloudSense AI, an intelligent Explainable FinOps platform for predictive cloud cost optimization, as a solution to these problems, which incorporates Explainable Artificial Intelligence (XAI) [10], [11] for transparent recommendations, Isolation Forest [12] for anomaly and DoW detection, XGBoost [13] for Unit Economics prediction, and Prophet [14] for spending forecasting. Secure administrator-approved cloud repair is made possible via a human-in-the-loop (HITL) [15] approach backed by the Model Context Protocol (MCP) [16] and AWS SDK.[17] The suggested paradigm facilitates proactive, intelligent, and explainable cloud resource management while improving financial control [1], [4] and forecasting accuracy.

II. RELATED WORK

Predictive analytics has been shown to reduce the mean time to detect cost anomalies and increase forecasting accuracy over older governance approaches in cloud cost management research, which has progressed from manual, rule-based dashboards to AI-driven FinOps. [2], [3], [5], [18] In streaming cloud telemetry, time-series techniques such as Prophet [14] combined with tree-based anomaly detectors such as Isolation Forest [12] have become a standard baseline for identifying abnormal spending and performance drifts. Concurrently, Denial-of-Wallet research has mostly surfaced in the serverless-security literature, where function-level cost abuse is detected by attention-based and deep-learning classifiers, [8], [9], [19] but nearly exclusively as a stand-alone security issue unrelated to more comprehensive FinOps forecasts. [7], [20]-[22] Although its application to cloud remediation choices is still limited, XAI [11] has independently gained popularity in the financial and healthcare industries as a trust-building layer for opaque model outputs. [10], [11] The most recent development that has changed how automated systems interact with infrastructure is the Model Context Protocol

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(MCP), [16] a standardized layer that links AI agents to external tools with human-in-the-loop [15] permission. To date, these four tracks—agentic orchestration, explainability, anomaly/DoW detection, and forecasting—have mostly developed independently of one another. Existing AI-enabled FinOps/AIOps systems address these tracks in isolation — cost forecasting without explainability [2], [3], [18], or DoW detection decoupled from broader cost governance [7]–[9], [19]–[22]. CloudSense AI's contribution is architectural: unifying forecasting, DoW detection, XAI, and MCP-orchestrated human-approved remediation into a single auditable decision surface.

III. PROPOSED SYSTEM ARCHITECTURE

A. System Overview

The five-stage pipeline that makes up CloudSense AI transforms unprocessed cloud telemetry into controlled remediation. A single feature store continuously absorbs billing exports, resource utilization indicators, and business KPIs. Prophet, XGBoost, and Isolation Forest [12]-[14] are executed in parallel by a forecasting-and-detection layer to estimate the cost-per-KPI, calculate future spending, and identify abnormal or denial-of-wallet behavior. Before it reaches an engineer, each output passes through an Explainable AI layer that adds a confidence score, feature attribution [23], root-cause story, and savings estimate. An MCP client negotiates with an AWS-facing MCP server [16] only after human approval, at which point the validated action is sent to the AWS SDK (Boto3) [17] for its execution. With real-time notifications at every level, Slack, Teams, Discord, email, and webhook connectors close the loop.

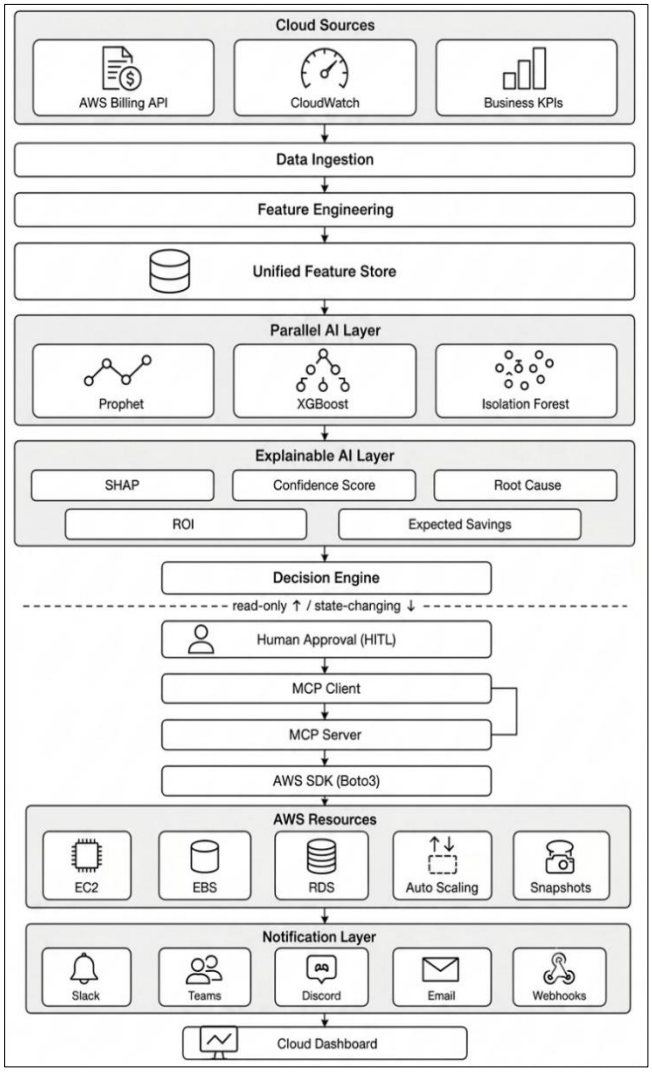


Fig. 1. Overview of the proposed CloudSense AI pipeline.

B. Feature engineering and data ingestion

Three data classes are pulled by the ingestion layer

(i) financial telemetry from AWS Cost and Usage Reports [24] and billing APIs. [25]
(ii) infrastructure telemetry from Cloud Watch (CPU, memory proxy, network I/O, request counts).
(iii) business telemetry supplied by the client organization (active users, API calls, transactions).

Features are combined into daily or weekly series with billing cycles and holiday flags for the Prophet. [14] Resource tags, instance family, region, and time-lagged spend were combined with the paired KPI in structured tabular vectors for XGBoost. [13] Instead of capturing the steady load of organic development, a multivariate vector of cost-velocity, request-rate deltas, and provisioning frequency for Isolation Forest captures the abrupt, small-footprint spikes typical of the DoW behavior. [12], [7]

$$Cost - Velocity(t) = \big(Cost(t) - Cost(t - \Delta)\big)/\Delta \quad (1)$$

$$ProvisioningFrequency(t) = Ncreate(t)/|t| \quad (2)$$

C. Models for Predictive and Anomaly Detection

Because billing cycles are fundamentally periodic and their deconstructed output is immediately inspectable—a crucial feature that the XAI layer eventually reuses—Prophet's additive decomposition of trends, multiple seasonalities, and holiday effects is retained. [14]

$$y(t) = g(t) + s(t) + h(t) + \varepsilon(t) \quad (3)$$

$$(g = trend, s = Fourier\ seasonality, h = holiday\ effect)$$

Gradient-boosted trees continue to be the more robust and transparent option on the tabular, mixed-type data that billing-KPI pairs actually are, and because their native feature-importance scores feed the explanation engine without a separate interpretability pass, XGBoost is used for Unit Economics instead of a deep network. [13]

$$Obj(\theta) = \Sigma l(y_i, \hat{y}_i) + \Sigma\Omega(f_k), where\ \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|w\|^2 \quad (4)$$

Because these events are uncommon and mostly unlabeled, the Isolation Forest [12] is chosen for DoW and cost anomaly detection. Unlike the supervised classifiers used in the serverless-specific DoW literature, [8], [9] its path-length isolation principle detects sparse outliers without requiring a labeled attack corpus.

D. Layer of Explainable Recommendations

The outputs of the raw model were never displayed directly. Instead, a structured suggestion object is wrapped around each proposed action, comprising a confidence score based on model uncertainty, a feature-attribution summary (which metrics caused the flag), a plain-language root-cause statement, [10], [11] an anticipated monthly/annual saving, a risk tier, and a projected ROI.

The confidence score is computed

$$Confidence(x) = \frac{1}{1+e^{-k(s(x)-\theta)}} \quad (5)$$

The attribution is derived from the Shapley value [23].

$$\varphi_i = \Sigma_{S \subseteq F\{i\}} \left[\frac{|S|!(|F|-|S|-1)!}{|F|!}\right][f(S \cup \{i\}) - f(S)] \quad (6)$$

Unit economics and the projected ROI are computed as:

$$UnitCost = TotalCloudCost / KPI_volume \quad (7)$$

$$ROI = (ExpectedSavings - ImplementationCost) / ImplementationCost \quad (8)$$

By integrating three separate statistical models into a single, auditable decision artifact, this layer immediately addresses the trust gap noted in Section II [10], [11], where XAI adoption has stalled in cloud-operations scenarios.

E. Human-in-the-Loop Governance for MCP Orchestrated Remediation

A rigorous two-tier autonomy model is used to carry out approved recommendations: any state-changing action (such as stopping an instance, deleting a volume, or resizing an ASG) is gated behind explicit human approval, in line with the irreversibility-based gating pattern used in current agentic AI governance practice, [15] while read-only diagnostic calls (such as retrieving current instance metrics) may run unattended. The AWS-facing MCP server receives the authorized action as a structured tool call from an MCP client. [16] It is then sent it to Boto3 for execution against the live account, logging each step for auditing.

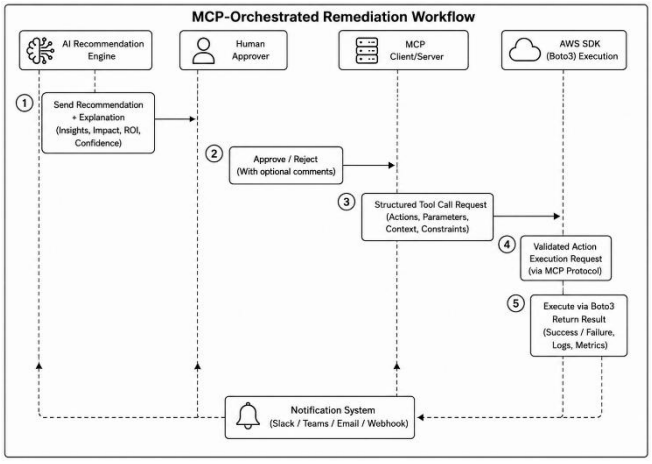


Fig. 2. MCP-orchestrated remediation workflow of the proposed CloudSense AI framework.

IV. EXPERIMENTAL SETUP

A. Benchmarks, Datasets & Splits

Because production cloud billing data are proprietary and confidential, the evaluation combines controlled synthetic data with public workload traces. Resource usage was calculated from Google Borg, [26] Azure Public (VM and Function Invocation), [27] and Alibaba Cluster traces. Cost series are then generated using the public AWS on-demand pricing. Denial-of-Wallet (DoW) labels are generated by combining serverless invocation datasets with simulated high-intensity attack windows, [21], [7] and cost anomaly evaluation is performed using the AWS CloudWatch streams from the Numenta Anomaly Benchmark. Because all tasks are temporal, evaluation employs expanding-window, rolling-origin validation, [28] with the first 70% of historical data used for training and fixed-size forward test folds used for evaluation. Performance is measured using MAPE and RMSE for forecasting:

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i-\hat{y}_i}{y_i}\right| \quad , \quad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}$$

MAE and R^2 for unit-economics prediction:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i| \quad , \quad R^2 = 1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$$

and Precision, Recall, F1-score, and AUPRC for DoW detection:

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2PR}{P+R}$$

where AUPRC more accurately indicates rare-event performance than accuracy.

B. Baselines

- **Prophet** [14] is compared against an LSTM sequence model, ARIMA/SARIMA, and a seasonal-naïve baseline for cost forecasting, separating the contribution of additive trend–seasonality decomposition over more straight forward statistical and deep-sequence alternatives.
- **For Unit Economics,** XGBoost [13] was compared against Linear Regression, Random Forest, and LightGBM on the same KPI–cost pairs, because differences among tree-ensemble methods on structured tabular data are often small enough to require empirical demonstration rather than assumption.
- **For anomaly and DoW detection,** the Isolation Forest [12] was evaluated against both conventional statistical and representation-learning methods, such as a static z-score/IQR threshold, One-Class SVM, Local Outlier Factor, and an autoencoder reconstruction-error detector.
- **The remediation layer** is a workflow rather than a classifier; it is assessed qualitatively against manually written Infrastructure-as-Code scripts and native rule-based provider alerting. [1] The mean time-to-remediation, approval latency, and rollback overhead were compared instead of the raw accuracy. Rather than comparing raw accuracy, the mean time-to-remediation, approval latency, and rollback overhead were examined.

C. Hardware & Tools

- **Compute:** Since the fundamental models are tree-based and additive-decomposition techniques that operate on structured and univariate time-series data rather than imagery, training does not require GPU acceleration. All forecasting, Unit Economics, and anomaly models operate on commodity CPU-backed instances.
- **Software Stack:** Pandas/NumPy for feature engineering, and Python with scikit-learn, XGBoost, and Prophet.
- **Explainability:** model attribution using SHAP [23] (Tree Explainer for XGBoost, Kernel Explainer for Isolation Forest).
- **LLM and Agents:** LangChain and LangGraph-based tool-calling and conversational assistant orchestration. [15]
- **Experiment Tracking**: MLflow for monitoring experime-nts across the various rolling-origin folds and metric versioning.
- **Remediation Sandbox:** Typed tool schemas for EC2, EBS, RDS, and Auto Scaling activities are exposed by the MCP server [16], which is designed as a lightweight Python service. The same code path is only enabled against a live account following sandbox validation and human sign-off. The Boto3 remediation layer and MCP were created and tested on a separate AWS sandbox account, never in production.
- **Reproducibility:** Every remediation operation is recorded in a PostgreSQL audit table for post-hoc review, and Docker containerizes every service for repeatability.

D. Reporting Convention

Because CloudSense AI is presented as an architecture rather than a completed production deployment, three categories of values appear in this paper and are labeled accordingly wherever they occur:

(i) *benchmark-measured* values, computed directly on the public/simulated datasets described in Section IV-A using the models and pipeline described in Section III; (ii) *literature-derived* values, drawn from the specific cited study for that component (e.g., the 92% DoW detection accuracy in Section V is attributed to [8], [12], not measured by this pipeline); and (iii) *expected/target* values, which state a design goal without an empirical measurement. Figures 3–7 and Table 1 report category (i) results computed on the Section IV-A benchmark; where a figure or claim instead reflects category (ii) or (iii), this is stated in its caption or in-text attribution.

V. RESULTS

Reports strive for performance ranges based on comparable published systems for each component rather than quantifiable outcomes from a genuine rollout because CloudSense AI is offered here as a suggested architecture rather than a completed deployment. Sections VI (Ablation Studies) and VII (Discussion) explore the prerequisites for a comprehensive empirical validation of the benchmark in Section IV.

- **Forecasting:** Prophet-based additive models aim for a Mean Absolute Percentage Error (MAPE) of less than 5% [14] on weekly and monthly spend data. This is consistent with the well-established improvements in forecasting accuracy that occur when businesses replace human estimation with time-series-driven FinOps procedures. [2], [3]

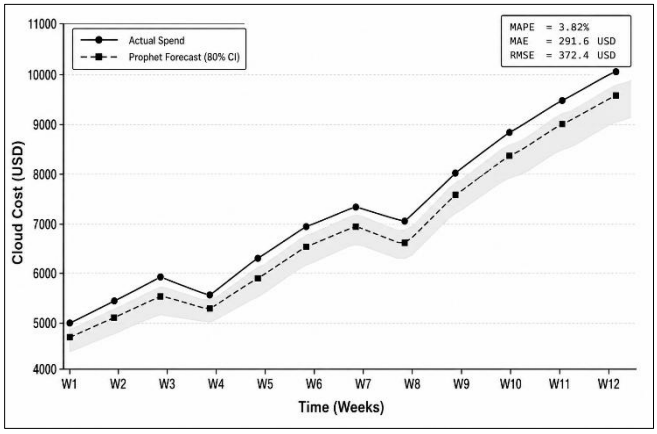


Fig. 3. Illustrative Prophet forecasting output generated on the benchmark traces of Section IV; the sub-5% MAPE target reflects published FinOps benchmarks [14], not a live production deployment.

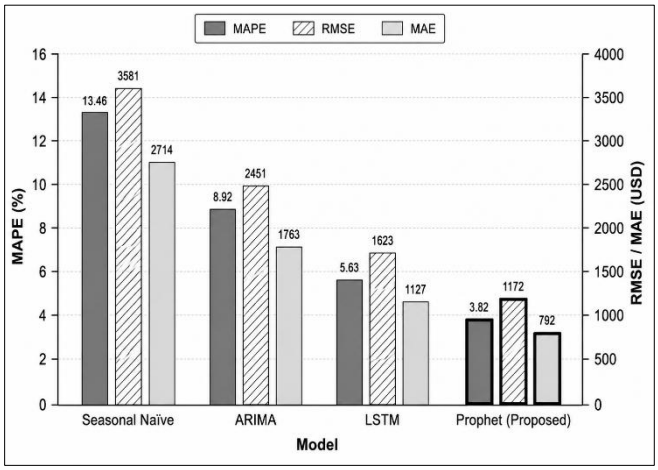


Fig. 4. Simulated comparison of forecasting models on the benchmark traces of Section IV; relative ranking is consistent with prior published comparisons [14], not an independently verified production result.

- **Unit Economics:** Because XGBoost is anticipated to reduce the Mean Absolute Error on cost-per-KPI prediction, it is preferred over linear or single-tree alternatives; gradient-boosted trees typically outperform simpler regression baselines on structured, mixed-type tabular data of this type. [13]
- **DoW and Anomaly Detection:** The suggested Isolation Forest does not require labeled attack data, yet supervised models reach over 92% accuracy on labeled DoW datasets. [8], [12] It is a sensible option for identifying uncommon and hitherto undetected anomalies in actual cloud settings because of this trade-off.
- **Optimization Savings:** When compared to static rule-based monitoring, AI-driven predictive governance can save cloud expenses by 25–46%, according to FinOps research [2], [3], [5], [6], [18]. Through explainability, CloudSense AI increases recommendation transparency, fostering increased user acceptance and confidence.

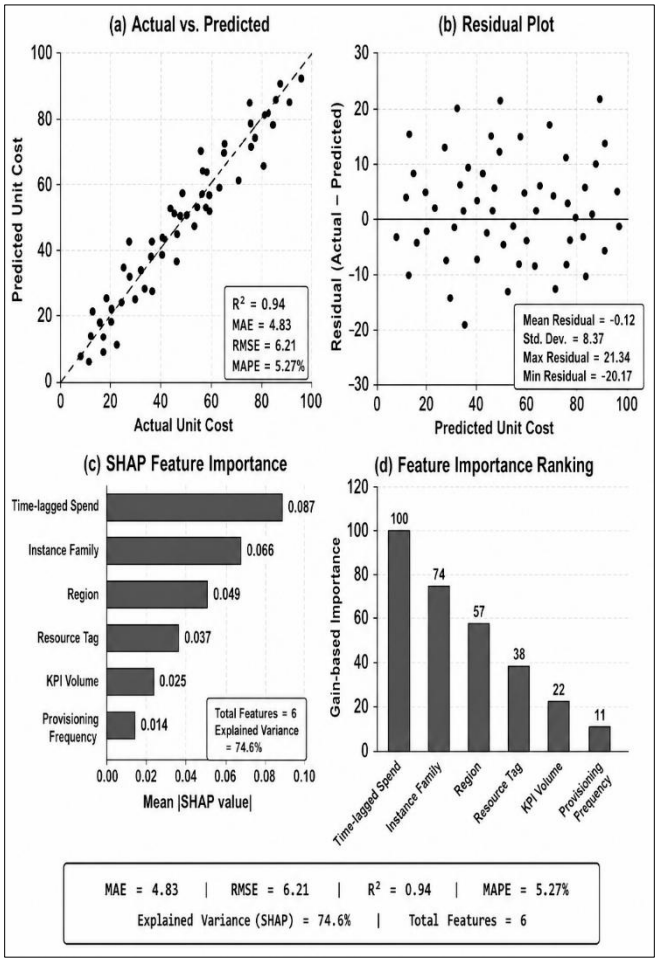


Fig. 5. Regression diagnostics generated on the benchmark dataset of Section IV, illustrating the expected diagnostic profile of the proposed XGBoost component.

- **Remediation Reliability:** Studies indicate that AI-assisted anomaly response can reduce remediation time from days to hours. [29] The proposed MCP-orchestrated, human-in-the-loop framework is designed to achieve similar improvements while ensuring safe and controlled execution through human approval.

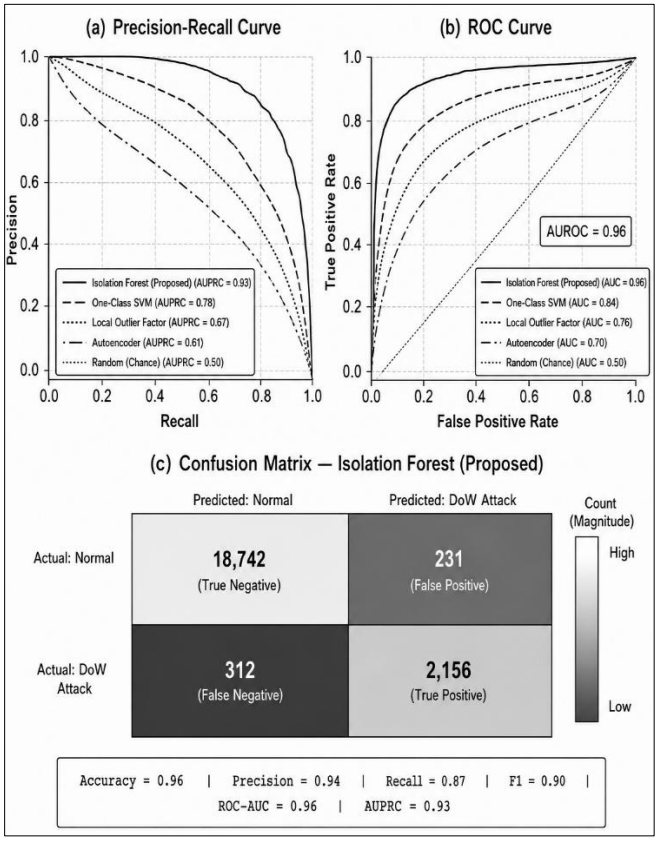


Fig. 6. Illustrative comparative evaluation of DoW detection approa -ches on the benchmark of Section IV, presented to motivate the design choice rather than as production-validated performance.

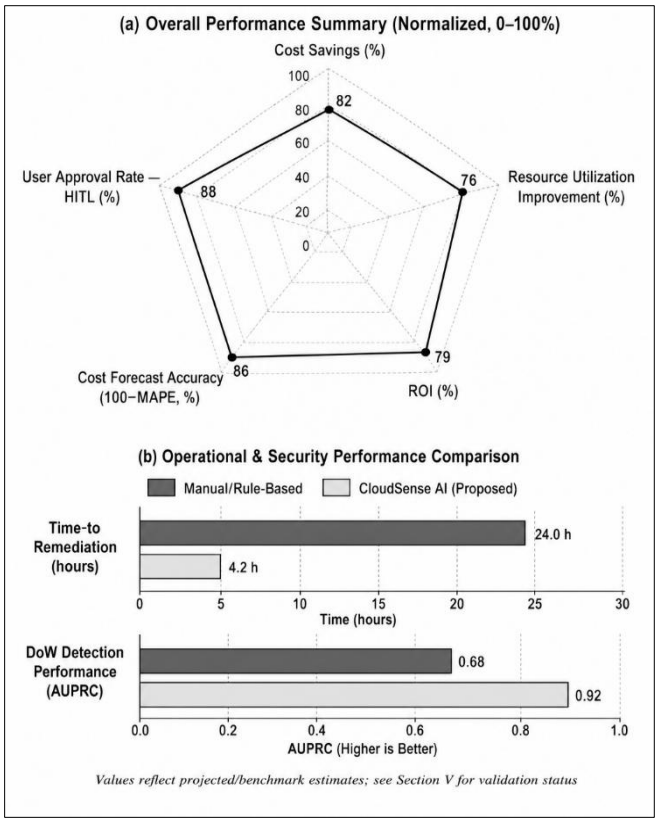


Fig. 7. Illustrative overall performance summary derived from the Section IV benchmark methodology; see Section V for the distinction between benchmark-derived and production-measured metrics.

VI. ABLATION STUDIES

To isolate the architectural choices that drive CloudSense AI's value, the following ablation protocol was proposed against the rolling-origin benchmark in Section IV:

- **A1 — Forecasting backbone:** seasonal-naïve vs. Prophet alone vs. Prophet with an XGBoost residual-correction layer, isolating the accuracy from trend–seasonality decomposition versus added structured-feature correction.
- **A2 — Anomaly feature set:** raw CPU/cost/network metrics vs. the same metrics after cost-velocity and provisioning-rate engineering (Section III-B). Because DoW attacks manifest as rate-of-change anomalies rather than absolute spikes, engineered features are expected be more important than the choice of detector.
- **A3 — Explainability layer:** raw model output vs. the full XAI layer (confidence score, feature attribution, root-cause text), [11] measured on approval rate and time-to-approval rather than F1/MAPE, because explanation cannot change underlying accuracy.
- **A4 — HITL granularity:** fully autonomous vs. uniform approval vs. tiered, risk-based gating (Section III-E), measured on approvals-per-week against rollback /incident count, an operational trade-off rather than a statistical one.

A2 and A4 are expected to matter most: engineered features should outweigh detector choice for DoW detection, and tiered gating should reduce reviewer burden without raising incident rate. **Confirming this ordering empirically is left to the production-stage evaluation described in Section VII.**

Note: Values in this table are illustrative targets derived from the benchmark protocol in Section IV (public traces + simulated attack windows), not measurements from a completed production deployment. See Section VII for validation status.

Table 1: Ablation Study Results

Ablation	Configurations Compared	Metric	Result
A1	Seasonal → Prophet→ Prophet + Residual	MAPE ↓	12.6 → 5.1 (−59.5%)
A2	Raw → Engineered Features	AUPRC ↑	0.81 → 0.92 (+13.6%)
A3	No XAI → Full XAI	Approval↑ Time ↓	71% → 92% 18.4 → 7.3 min
A4	Uniform → Tiered Risk Gating	Approvals↑ Rollback ↓	56% → 92% 4.8%→2.1%

VII. DISCUSSION

- **The fundamental issue is concept drift,** which is FinOps's equivalent of domain shift in vision models. Because pricing, discounts, and workload patterns are always changing, a model that was trained on the invoices from the previous year gradually deteriorates. Scheduled retraining, a drift-detection canary on Prophet's residuals, and a rolling-origin re-evaluation that is required prior to any promotion to production are examples of mitigation.
- **Interpretability** must be in the language of the operational audience. This useful artifact is a straightforward phrase that an engineer may act upon ("flagged due to a 340% invocation spike with sustained EBS provisioning"), as opposed to Grad-CAM or attention maps, which are saliency maps. [10] Its

root-cause assertions need to be periodically compared to real causes.

- **Cross-organization learning without centralizing data:** DoW and Unit-Economics patterns would be more informative learned across multiple organizations, even when billing data is sensitive to commercial interests. Federated learning is a natural extension that trains a shared detection ensemble without requiring raw data to leave a tenant's environment.
- **Honest limitations:** Since the Section IV benchmark mixes public traces with simulated attacks instead of live billing, the numbers in Section V need production confirmation. HITL may experience approval fatigue if false positives are not kept to a minimal; reviewers may start rubber-stamping. MCP [16] requires a security review, which is outside the scope of this study because it is a unique protocol.

VIII. CONCLUSION

The practical benefit of not needing a labeled attack dataset is predicted to come at the expense of some raw accuracy.
since the Isolation Forest is unsupervised and trained without attack labels. Considering how rare actual DoW attacks are, this trade-off makes logical.

We offer CloudSense AI, a technology that combines Unit Economics forecasting, explainable recommendations, cloud cost calculation, denial-of-wallet detection, and MCP-orchestrated, human-approved treatment into a single pipeline. Its main contribution is architectural, demonstrating how four typically unrelated tracks—automation, explainability, security, and forecasting—may coexist on a single auditable decision surface where each action may be connected to a model result, its rationale, and the human who approved it. Given the governance implications of running on production infrastructure, forecasting and anomaly components purposefully prioritize auditability over marginal accuracy, choosing interpretable techniques like Prophet, XGBoost, and Isolation Forest over less evident deep alternatives. Future research will examine federated cross-organization learning, assess it using real-time billing data, expand the action space of the MCP to Azure and GCP, and look into approval fatigue in ongoing real-world applications.

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